Curvature Polarization Transport Gravity: A Variational Framework for Galaxies and Cluster Mergers

Carter L Glass Jr

Abstract

We present Curvature Polarization Transport Gravity (CPTG), a unified geometric theory in which baryonic matter sources a nonlinear curvature response governed by a single action-based framework. The theory is built around two coupled effects: curvature polarization, which modifies the gravitational response through the local invariant structure of the field, and curvature transport, which allows organized curvature to be directionally redistributed in dynamical systems. In galaxies, the dominant manifestation is a polarization-supported enhancement of the gravitational response that naturally yields flat rotation curves without invoking dark matter. In merging galaxy clusters, the same framework predicts that curvature need not remain pointwise coincident with the collisional baryonic plasma: broad polarization support and directional transport together generate displaced lensing structure without requiring an added collisionless mass component. Unlike approaches that rely on fixed empirical interpolation laws or additional matter fields, CPTG derives both regimes from one nonlinear curvature law tied to structural and cosmological scales. The result is a single theory with a Newtonian limit, a polarization-dominated galaxy regime, and a polarization-plus-transport merger regime, providing a testable alternative to both dark matter and phenomenological modified gravity models.

1 Introduction

Observed galaxy rotation curves [3] and gravitational lensing in merging galaxy clusters [4] exhibit persistent discrepancies when interpreted using baryonic matter alone in standard Newtonian or relativistic gravity. The conventional response is to enlarge the matter sector by introducing a non-baryonic dark component [2, 5]. An alternative is to ask whether one geometric framework can account for both the galaxy and cluster regimes by modifying the gravitational response itself.

In this work we develop *Curvature Polarization Transport Gravity* (CPTG) [11], an action-based nonlinear theory in which spacetime curvature depends not only on local field strength but also on the structural properties of the field, including gradient content and directional organization. The central claim is not that gravity is simply made stronger by hand, but that curvature can respond through two related mechanisms: curvature polarization and curvature transport. Polarization supplies a broad nonlinear enhancement of the response to baryonic structure, while transport governs how organized curvature can be redistributed anisotropically when the system is dynamical.

This distinction is essential for the phenomenology developed below. In quasi-static bound systems such as disk galaxies, the dominant effect is curvature polarization, yielding an enhanced effective response without introducing unseen matter. In merger environments, the directional organization and transport of curvature become explicit, allowing the observable lensing structure to separate from the collisional gas while remaining part of the same underlying geometric framework. CPTG is therefore presented here not as separate galaxy and cluster constructions, but as one theory whose Newtonian limit, polarization-dominated galaxy regime, and polarization-plus-transport merger regime arise from the same action.

2 Master Action

The CPTG framework is defined by the action

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + \mathcal{L}_m + \beta a_\star^{2/3} C^{1/3} + \mathcal{L}_{\text{trans}} \right], \tag{1}$$

where R is the Ricci scalar and the term $\frac{R}{16\pi G}$ corresponds to the Einstein–Hilbert action [10], with G Newton's constant, $\mathcal{L}_m$ is the matter Lagrangian, β is the polarization coupling, C is the curvature-polarization invariant, and $\mathcal{L}_{\text{trans}}$ is the transport-sector Lagrangian.

Here β carries dimensions such that the polarization term has units of energy density. The scale $a_\star$ enters as a characteristic acceleration associated with the polarization sector; in galaxy-scale applications it is inferred from the structure of each system rather than treated as a universal constant.

The additional CPTG terms modify the gravitational response without introducing a dark matter component. The term $\beta a_\star^{2/3} C^{1/3}$ controls nonlinear curvature polarization, while $\mathcal{L}_{\text{trans}}$ governs directional redistribution of curvature. Although the invariant C governs the polarization dynamics at the level of the action, its detailed dependence does not appear explicitly in the reduced galaxy-limit equation, where its effects are encoded in the effective nonlinear structure.

3 Field Definitions and Baseline Gravity

We define the gravitational field and its magnitude by

$$\mathbf{g} \equiv -\nabla\Phi, \qquad g \equiv |\mathbf{g}|, \tag{2}$$

where Φ is the gravitational potential.

The standard Poisson equation is

$$\nabla^2\Phi = 4\pi G\rho, \tag{3}$$

where ρ is the baryonic mass density. Therefore,

$$\nabla \cdot \mathbf{g} = -4\pi G\rho. \tag{4}$$

4 Curvature Invariant

In reduced spatial component form, the CPTG curvature invariant is

$$C = g_j g_j + \kappa^2(\partial_i g_j)(\partial_i g_j), \tag{5}$$

where κ is the curvature-gradient length scale.

The invariant contains two contributions:

- the local field term $g_j g_j$,
- the gradient-structure term $\kappa^2(\partial_i g_j)(\partial_i g_j)$.

The first measures the local gravitational strength, while the second measures spatial structure. Together they define the nonlinear response in the polarization sector at the level of the action. The parameter κ sets the characteristic length scale over which spatial variations of the gravitational field contribute to this response.

In the reduced galaxy-limit equation, the detailed dependence on C does not appear explicitly; its effects are encoded in the effective nonlinear structure of the field equation.

5 Variation of the Polarization Sector

The polarization action is

$$S_P = \int d^4x \sqrt{-g}\, \beta a_\star^{2/3} C^{1/3}. \tag{6}$$

The invariant is

$$C = g_j g_j + \kappa^2 (\partial_i g_j)(\partial_i g_j). \tag{7}$$

Since $\mathbf{g} = -\nabla\Phi$, the variation must be taken with respect to Φ.

We compute

$$\delta g_j = -\partial_j(\delta\Phi), \qquad \delta g = \frac{g_j}{g}\,\delta g_j. \tag{8}$$

This expression for δg is understood away from isolated points where $g = 0$.

The variation of C becomes

$$\delta C = 2g_j\,\delta g_j + 2\kappa^2 (\partial_i g_j)(\partial_i \delta g_j). \tag{9}$$

Substituting,

$$\delta C = -2g_j\,\partial_j(\delta\Phi) - 2\kappa^2 (\partial_i g_j)(\partial_i \partial_j \delta\Phi). \tag{10}$$

The variation of the action is

$$\delta S_P = \int d^4x \sqrt{-g}\,\beta a_\star^{2/3} \frac{1}{3} C^{-2/3}\,\delta C. \tag{11}$$

Integrating by parts (twice for the second term), and retaining only the leading divergence contribution while neglecting higher-order derivative couplings as part of the coarse-grained reduction to the galaxy-limit equation, yields the Euler–Lagrange contribution to the field equation:

$$\partial_j \left[\frac{\beta a_\star^{2/3}}{3} C^{-2/3} \left(g_j - \kappa^2 \partial_i \partial_i g_j \right) \right]. \tag{12}$$

This contribution enters the full field equation with the prefactor $\frac{\beta a_\star^{2/3}}{3}$. In the reduced galaxy limit, the polarization coupling β is absorbed into the normalization of the effective nonlinear response and does not appear as an independent parameter in the galaxy-scale equation.

6 Transport Sector

We define

$$\mathcal{L}_{\text{trans}} = -\frac{\lambda_t}{2} \left(S^\alpha \Sigma \nabla_\alpha \Phi \right)^2, \tag{13}$$

where λ_t is the transport coupling.

The transport direction field S^α is taken to be a unit vector, satisfying $S^\alpha S_\alpha = 1$, and is treated as fixed under variation, representing the dominant direction of curvature transport in the reduced theory.

$$\Sigma = \left(1 + \left(\frac{g}{a_\star}\right)^{123/50}\right)^{-11/50} \tag{14}$$

This expression is understood as an effective suppression factor in the reduced theory and is defined away from isolated points where $g = 0$.

$$a_\star = \frac{3}{16\pi} c H_0 \tag{15}$$

Define

$$\Psi = S^\alpha \Sigma \nabla_\alpha \Phi. \tag{16}$$

Then

$$\delta \mathcal{L}_{\text{trans}} = -\lambda_t \Psi \, \delta\Psi. \tag{17}$$

$$\delta\Sigma = \frac{\partial \Sigma}{\partial g} \, \delta g. \tag{18}$$

The variation is

$$\delta\Psi = S^\alpha \left[(\delta\Sigma)\nabla_\alpha \Phi + \Sigma \nabla_\alpha (\delta\Phi)\right]. \tag{19}$$

The second term yields (after integration by parts):

$$\nabla_\alpha \left(\lambda_t S^\alpha S^\beta \Sigma^2 \nabla_\beta \Phi\right). \tag{20}$$

The first term produces additional nonlinear corrections through $\delta\Sigma = \frac{\partial\Sigma}{\partial g}\,\delta g$. In the reduced effective theory considered here, the $\delta\Sigma$ contribution is subleading in the quasi-static regime and is neglected to isolate the dominant transport contribution.

The leading transport operator in the reduced theory is

$$\nabla_\alpha \left(\lambda_t S^\alpha S^\beta \Sigma^2 \nabla_\beta \Phi\right). \tag{21}$$

In the reduced spatial formulation used for galaxy and lensing applications, this operator is projected into divergence form.

In the reduced galaxy limit, the transport coupling λ_t is absorbed into the normalization of the effective geometric term and does not appear as an independent parameter in the galaxy-scale equation.

7 Full CPTG Field Equation

Here the equation is expressed in a mixed representation: the polarization contribution is written in reduced spatial component form, while the transport contribution is retained in its covariant form prior to projection into the reduced description. The governing equation is:

$$\nabla \cdot \mathbf{g} = -4\pi G\rho + \nabla \cdot \left[\frac{\beta a_\star^{2/3}}{3} C^{-2/3} \left(\mathbf{g} - \kappa^2 \nabla^2 \mathbf{g} \right) \right] + \nabla \cdot \left(\lambda_t \, \Sigma^2 \, \mathbf{S} \, (\mathbf{S} \cdot \nabla) \Phi \right). \quad (22)$$

7.1 Relation to General Relativity

CPTG is formulated as an extension of the Einstein–Hilbert action [10, 1] rather than as a replacement for it. The standard relativistic curvature term,

$$\frac{R}{16\pi G},$$

is retained in the master action, so that the theory preserves the usual geometric interpretation of gravity as spacetime curvature sourced by matter. The additional CPTG terms introduce nonlinear curvature-polarization and transport responses that become relevant in low-acceleration or dynamically organized regimes.

In the high-acceleration limit, and whenever the polarization and transport contributions are subdominant, the theory reduces to the ordinary gravitational response. In the weak-field quasi-static limit this corresponds to the standard Newtonian Poisson equation,

$$\nabla^2 \Phi = 4\pi G\rho.$$

Thus CPTG is constructed to recover the usual Einstein–Newtonian behavior in the appropriate limit, while modifying the effective response in regimes where the curvature invariant and transport structure become dynamically important.

The galaxy and cluster equations used in this paper should therefore be understood as reduced weak-field and effective-sector limits of the broader action-based framework. A fully covariant treatment of the complete nonlinear field equations is beyond the scope of the present work and is left for future development.

8 Parameter List

The principal parameters and variables of CPTG are:

- **Fundamental constants and fields**
  - c: speed of light,
  - G: Newton's gravitational constant,

- ρ: baryonic mass density,
- Φ: gravitational potential,
- $\mathbf{g} = -\nabla\Phi, \quad g = |\mathbf{g}|$: gravitational field and magnitude.

- **Full CPTG theory**
  - β: polarization coupling appearing in the action,
  - λ_t: transport coupling appearing in the action,
  - S^α: covariant transport direction field in the full theory.

- **Galaxy-limit quantities**
  - $a_\star$: characteristic acceleration scale, structurally derived per system,
  - H_0: structure-dependent expansion scale related to $a_\star$ through

$$H_0 = \frac{16\pi}{3}\frac{a_\star}{c},$$

  - R_c: derived structural transport scale,

$$R_c = \frac{48\pi c^2}{a_\star},$$

  - η: dimensionless geometric transport coefficient fixed by the reduced galaxy-limit geometry.

- **Reduced cluster-merger quantities**
  - $\mathbf{S}$: effective spatial transport direction in the reduced lensing description,
  - s: transport strength in the reduced lensing equation,
  - Σ: nonlinear suppression factor,
  - $\chi_{\rm bg}$: background curvature-organization field that guides transport,
  - $\mathbf{u}_{\rm eff}$: effective reduced transport field,
  - T_+: positive transported-curvature mode,
  - C_T: local transported-curvature invariant,
  - $\Sigma_{\rm pol}$: static polarization-supported lensing background,
  - Σ_T: transported-curvature lensing contribution,
  - $\alpha_{T,\rm eff}$: effective transported-curvature weighting.

All terms in the effective galaxy equation carry dimensions of acceleration, with the denominator dimensionless, while $\mathcal{K}_{\rm lens}$ carries the dimensions appropriate to the effective lensing-curvature sector.

9 Physical Interpretation and Conceptual Overview

While the formal development of Curvature Polarization Transport Gravity (CPTG) is expressed through a nonlinear variational framework, its physical content can be stated more directly. CPTG is a single theory in which baryonic matter sources curvature, the resulting curvature develops a nonlinear polarized response, and organized curvature can also be transported anisotropically. These are not separate additions to gravity, but coupled consequences of the same action and the same invariant structure.

Accordingly, the gravitational response in CPTG depends not only on local mass density, but also on the strength, gradients, and directional organization of the field itself. The two principal physical mechanisms are curvature polarization and curvature transport.

9.1 Curvature Polarization

The polarization sector describes the way curvature responds nonlinearly to its own local structure. Through the invariant C, the effective gravitational response depends not only on field magnitude but also on gradient content, so the field need not remain a strictly local or purely linear reflection of the baryonic source.

Physically, curvature polarization is a broad self-adjustment of the gravitational response. In low-acceleration environments, such as the outer regions of galaxies, this produces an enhanced effective gravitational field without introducing an additional matter component. In this sense, the observed excess gravitational response is interpreted as a curvature-supported amplification rather than as evidence for unseen mass.

In the high-acceleration limit, the polarization contribution becomes subdominant and the theory smoothly approaches the standard Newtonian regime. CPTG therefore preserves established strong-field behavior while allowing a controlled nonlinear departure in the low-acceleration domain.

9.2 Curvature Transport

The transport sector describes a different physical effect. Whereas polarization changes the *strength* of the local curvature response, transport governs the *organization and displacement* of curvature in space. This becomes especially important in dynamically evolving systems such as cluster mergers, where the relevant geometry is anisotropic and time dependent.

The transport term allows curvature to be guided along preferred directions defined by the vector field S^α. As a result, the effective lensing structure need not remain pointwise coincident with the collisional baryonic plasma. Instead, curvature can be redistributed and persist in displaced organized structures during merger dynamics.

This provides a geometric interpretation of cluster observations in which the lensing morphology is offset from the hot gas component. In CPTG, such offsets are attributed not to an added collisionless dark sector, but to transported curvature supported by the same nonlinear framework that governs polarization.

9.3 Unified Picture

Taken together, curvature polarization and transport define a single physical picture:

- Polarization provides the broad nonlinear enhancement of the gravitational response.
- Transport organizes and redistributes curvature anisotropically when the system evolves dynamically.
- The observed phenomenology depends on the relative importance of these two effects in a given regime.

In quasi-static systems such as galaxies, the dominant manifestation is the polarization-supported enhancement of the effective field, with transport entering through the structural organization of the solution. In cluster mergers, the same theory permits organized curvature to be displaced and maintained away from the collisional gas distribution. The central claim of CPTG is therefore not merely that gravity is strengthened, but that curvature can both reorganize its local response and develop directed large-scale structure within one action-based framework.

9.4 Summary

In conceptual terms, CPTG replaces the notion of gravity as a fixed response to mass with a more general picture in which curvature can both *adjust* through polarization and *reorganize and move* through transport. This yields a unified geometric framework for galaxy-scale gravitational enhancement and cluster-scale lensing offsets without introducing an additional dark matter component.

10 Galaxy Limit from the Field Equation (Sketch)

In the quasi-static, weak-field, and approximately axisymmetric limit relevant to rotationally supported galaxies, explicit time dependence is neglected and the field is described by a scalar potential $\Phi(r)$. In this regime the dominant CPTG effect is the nonlinear polarization response, while higher-order derivative couplings are suppressed in the coarse-grained description.

The transport sector does not disappear in the galaxy limit, but its directional structure no longer appears as an explicit displaced-curvature effect of the kind relevant to cluster mergers. Instead, after symmetry reduction and averaging, its leading contribution enters as an effective geometric correction to the radial field. Under these assumptions, the full CPTG field equation reduces to a scalar nonlinear relation for the radial acceleration field. The effective galaxy equation given in Eq. (24) should therefore be understood as the leading bound-system reduction of the same unified polarization–transport framework.

The nonlinear galaxy response ultimately arises from the curvature-dependent structure of the theory, which in the full formulation is governed by invariants of the form

$$C = g^2 + \kappa^2(\nabla g)^2. \tag{23}$$

In the galaxy limit, this detailed curvature-gradient dependence is not carried forward as a separate explicit modulation factor. Rather, its net effect is encoded in the effective nonlinear structure of the reduced equation itself, including the suppression law and the transport-induced geometric term.

11 Galaxy Limit (Effective Reduction)

The galaxy equation below is a coarse-grained effective reduction of the full CPTG field equation under quasi-static, bound-system, and approximate symmetry assumptions. It is not presented as a closed-form derivation from the full action. Rather, it is the effective galaxy-regime relation that retains the leading nonlinear CPTG response once the full curvature dynamics are reduced to a radial description.

In this regime, the dominant effect is curvature polarization, while directional transport survives only through an averaged geometric correction. The resulting bound-system field equation takes the implicit nonlinear form

$$g(r) = g_{\rm bar}(r) + \frac{\sqrt{a_\star\, g_{\rm bar}(r)} + g_{\rm geom}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}. \tag{24}$$

Here the square-root enhancement captures the emergent low-acceleration polarization response, while the denominator suppresses that enhancement back toward the baryonic limit in the inner high-field regime. No separate curvature-modulation factor is introduced beyond this structure; the effective nonlinearity is already contained in the implicit dependence on $g(r)$ and in the transport-induced geometric term.

The geometric correction term is given by

$$g_{\mathrm{geom}}(r) = \eta\, a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \tag{25}$$

where $\eta = 1/2$ is fixed by transport-sector averaging in the galaxy limit. In physical terms, g_{geom} is the residual galaxy-scale imprint of curvature transport after the directional transport sector has been coarse-grained into an effective radial contribution.

The structural transport scale is

$$R_c = \frac{c}{H_0}(16\pi)^2, \tag{26}$$

with

$$H_0 = \frac{16\pi}{3}\frac{a_\star}{c}. \tag{27}$$

The baryonic acceleration is computed from fixed CPTG-weighted velocity components,

$$g_{\mathrm{bar}}(r) = \frac{\frac{27}{50}v_{\mathrm{gas}}^2(r) + \frac{23}{50}v_{\mathrm{disk}}^2(r) + \frac{16}{25}v_{\mathrm{bul}}^2(r)}{r}. \tag{28}$$

Within this effective reduction, $g_{\mathrm{bar}}(r)$ is not merely a Newtonian bookkeeping term. It is the baryonic input to the nonlinear CPTG response, which is then reshaped by polarization and by the coarse-grained transport remnant encoded in $g_{\mathrm{geom}}(r)$.

The effective equation (24) is solved numerically as an implicit nonlinear relation for $g(r) > 0$. Its implicit structure is essential: the acceleration appears on both sides, and the outer scaling is not imposed by hand but emerges as a self-consistent fixed point of the same unified nonlinear dynamics.

11.1 Fixed Baryonic Source Kernel

In the CPTG benchmark, the baryonic source weights are not treated as adjustable nuisance parameters. Variants with altered gas, disk, and bulge weights were tested, including nearby denominator-50 alternatives and closure-preserving gas–disk splits. Across these tests, the benchmark preferred the fixed CPTG source kernel

$$w_{\mathrm{gas}} = \frac{27}{50}, \qquad w_{\mathrm{disk}} = \frac{23}{50}, \qquad w_{\mathrm{bulge}} = \frac{16}{25}. \tag{29}$$

The gas and disk coefficients form a closed extended-source pair:

$$w_{\mathrm{gas}} + w_{\mathrm{disk}} = \frac{27}{50} + \frac{23}{50} = 1. \tag{30}$$

Equivalently, the pair may be written as a symmetric split around 25/50:

$$w_{\mathrm{gas}} = \frac{25+2}{50}, \qquad w_{\mathrm{disk}} = \frac{25-2}{50}. \tag{31}$$

Thus the extended baryonic source is not freely rescaled; rather, it is partitioned into a slightly gas-favored closed source decomposition.

The bulge coefficient is structurally distinct from the gas–disk pair:

$$w_{\text{bulge}} = \frac{16}{25} = \left(\frac{4}{5}\right)^2. \tag{32}$$

Since the reduced CPTG low-acceleration response contains a square-root source dependence, this corresponds to a compact-source response amplitude

$$\sqrt{w_{\text{bulge}}} = \frac{4}{5}. \tag{33}$$

The preference for the fixed weights therefore suggests that CPTG performance is tied to a specific morphology-aware baryonic source decomposition. The gas and disk terms define a closed extended-source balance, while the bulge term supplies a separate compact-source response. In this interpretation, the benchmark preference is not evidence for freely tuned baryonic rescaling, but for a fixed structural source kernel within the reduced CPTG galaxy-limit model.

Component	Weight	Structural role
Gas	27/50	Extended source, slightly enhanced
Disk	23/50	Extended source, complementary to gas
Bulge	16/25 = 32/50	Compact-source response kernel

12 Limiting Behavior

The equation reproduces the expected limits.

For high acceleration,

$$g \gg a_\star \quad \Rightarrow \quad g \approx g_{\text{bar}}. \tag{34}$$

For low acceleration,

$$g \ll a_\star \quad \Rightarrow \quad g \sim \sqrt{a_\star g_{\text{bar}}} + g_{\text{geom}}(r). \tag{35}$$

In this regime, the nonlinear suppression term approaches unity, and the acceleration is governed by the combined polarization and transport contributions.

The characteristic structural scale is

$$\ell_s \sim \frac{g}{|\nabla g|}, \tag{36}$$

which measures the local variation scale of the field.

13 Cluster Lensing Regime (Effective Form)

The lensing equation below is the effective transport-retaining form used for cluster-merger applications. At this stage it should be understood as the implementation-level reduction of the transport sector, rather than as the fully expanded relativistic field equation in closed form.

In dynamically evolving systems such as merging clusters, the transport contribution must be retained explicitly. The effective lensing curvature is written as

$$\mathcal{K}_{\rm lens} = \nabla^2\Phi - s\,\nabla\cdot\left(\Sigma^2\,\mathbf{S}\,(\mathbf{S}\cdot\nabla)\Phi\right), \tag{37}$$

This expression is a reduced spatial projection of the full transport operator $\nabla_\alpha(\lambda_t S^\alpha S^\beta \Sigma^2 \nabla_\beta \Phi)$, in which the effective coupling absorbs normalization factors while preserving the quadratic dependence on Σ and the transport tensor is projected onto the leading spatial direction $\mathbf{S}$. The resulting form retains the dominant directional transport contribution in the reduced lensing description.

Here $\mathbf{S}$ denotes the effective spatial transport direction in the reduced lensing description.

The first term corresponds to the baryonic mass distribution through $\nabla^2\Phi = 4\pi G\rho$.

The quantity $\mathcal{K}_{\rm lens}$ is proportional to the observable weak-lensing convergence and represents the effective projected curvature in the lensing sector, where s is the transport strength in the lensing sector.

The second term redistributes curvature along the transport direction $\mathbf{S}$. This allows spatial separation between the gas distribution and lensing peaks during cluster mergers.

Equation (37) is the implementation-ready CPTG lensing equation for cluster mergers.

14 CPTG Outer Regime Statement

14.1 Asymptotic Behavior of Galaxy Rotation Curves

Within the Curvature Polarization Transport Gravity (CPTG) framework, the galaxy-scale field equation derived from the master action predicts a universal outer regime in the asymptotic low-acceleration limit of the extended solution. This regime can be probed by extending the model solution beyond the observed radial extent of galaxy rotation curves.

We define the local logarithmic slope of the rotation curve as

$$s(r) \equiv \frac{d\ln V}{d\ln r}. \tag{38}$$

To isolate the asymptotic behavior, we evaluate the CPTG solution at radii $r > R_{\max}$, where $R_{\max}$ is the outermost measured radius of a given galaxy. The baryonic source terms are held fixed at their outermost observed values (i.e., extrapolated as constant beyond $R_{\max}$), ensuring that the resulting behavior reflects the intrinsic gravitational response rather than additional structural assumptions.

14.2 Numerical Convergence to the Asymptotic Slope

Across the extended CPTG solutions, the local logarithmic slope develops a stable outer-regime trend beyond the observed rotation-curve domain. In the asymptotic extension test, each galaxy is continued beyond its outermost measured radius using the fixed-outer-source continuation rule, and the median local slope is evaluated over the stacked sample. The raw median slope rises from approximately

$$s(r \simeq 4R_{\max}) \approx 0.227 \tag{39}$$

to approximately

$$s(r \simeq 128R_{\max}) \approx 0.246, \tag{40}$$

while remaining below the theoretical limiting value over the finite plotted range.

The outer trend is well described by an asymptotic convergence form,

$$s(r) = s_\infty - \frac{A}{(r/R_{\max} + B)^p}, \tag{41}$$

with a free-asymptote fit giving

$$s_\infty \approx 0.250168. \tag{42}$$

This fit is used only to quantify the limiting trend; the figure below plots the raw median local slope rather than the fitted curve. This is consistent with the theoretical CPTG attractor

$$s_\infty = \frac{1}{4}. \tag{43}$$

This value follows from the asymptotic scaling structure of the CPTG galaxy equation under the fixed-outer-source continuation rule. In the extended solution, the outer baryonic velocity components are held fixed beyond $R_{\max}$, so the effective baryonic source scales as $g_{\rm bar} \propto r^{-1}$. In the low-acceleration regime, the square-root polarization response then gives $\sqrt{a_\star g_{\rm bar}} \propto r^{-1/2}$, producing $g \propto r^{-1/2}$ and therefore $V(r) \propto r^{1/4}$. The geometric transport remnant remains part of the full solution, but the leading $s = 1/4$ scaling follows from the low-acceleration polarization response under this continuation rule.

This result characterizes the asymptotic behavior of the extended effective solution of the CPTG galaxy equation, rather than the directly observed slope within current rotation-curve data.

14.3 Outer Regime Scaling Law

The asymptotic slope corresponds to a power-law behavior of the rotation curve,

$$V(r) \propto r^{1/4}, \tag{44}$$

and the corresponding gravitational acceleration scales as

$$g(r) \propto r^{-1/2}. \tag{45}$$

This defines the asymptotic *CPTG outer regime* of the extended effective solution, which emerges from the low-acceleration nonlinear structure of the CPTG galaxy equation under the fixed-outer-source continuation rule.

14.4 Relation to Observed Slopes

Observed outer slopes, measured over the final data points of rotation curves, are typically smaller:

$$\langle s_{\mathrm{obs}} \rangle \sim 0.05\text{–}0.15. \tag{46}$$

This discrepancy is explained by the fact that current observations predominantly sample a *transition region* between baryon-dominated dynamics and the asymptotic CPTG regime. As a result, the true outer attractor is not directly observed in most galaxies.

14.5 Visualization of Convergence

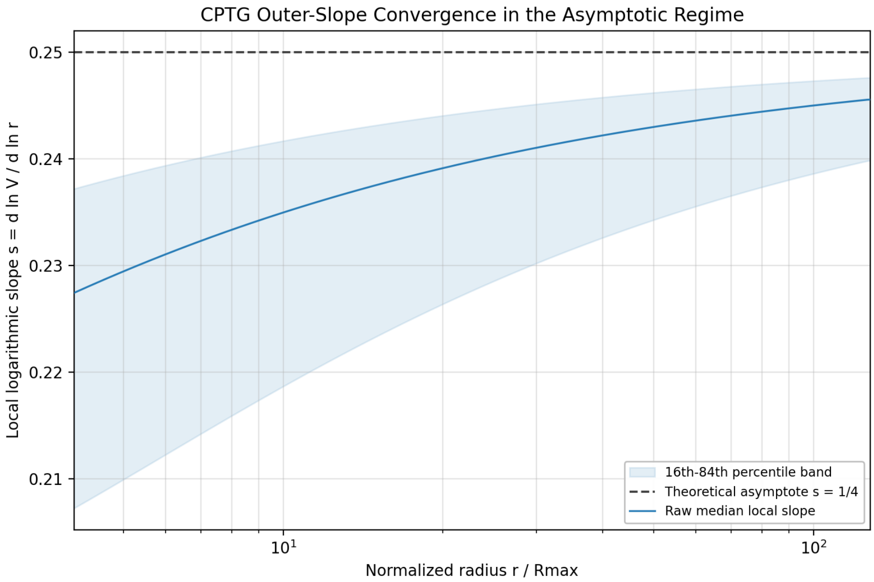

Figure 1: Raw median CPTG outer-slope convergence in the asymptotic extension domain, shown as a function of normalized radius r/R_{max} for the SPARC galaxy sample [8]. The solid curve gives the median local logarithmic slope $s = d\ln V/d\ln r$ computed from the CPTG-predicted extended rotation curves under the fixed-outer-source continuation rule. The shaded band shows the 16th–84th percentile range across the sample. The dashed line marks the theoretical CPTG outer-regime attractor $s = 1/4$. The upward approach toward this value indicates a low-acceleration asymptotic trend of the effective CPTG galaxy equation, rather than a directly observed slope beyond the present rotation-curve data.

This behavior shows that the CPTG extended solutions approach a stable outer-regime trend rather than drifting arbitrarily once extrapolated beyond the measured rotation-curve domain. Under the fixed-outer-source continuation rule, the stacked local slopes rise toward the low-acceleration $s = 1/4$ attractor, with the percentile band showing the remaining galaxy-to-galaxy spread.

14.6 Implications

The existence of a universal outer attractor has several important consequences:

1. The CPTG galaxy equation requires no additional correction terms in the outer regime.
2. The apparent variation in observed outer slopes arises from incomplete radial sampling rather than a breakdown of the theory.
3. The $s = 1/4$ scaling provides a direct, testable prediction for future deep observations of galaxy outskirts.

15 Structural Mode N

To characterize the internal organization of a bound system in the CPTG galaxy limit, we introduce a dimensionless structural mode

$$N \equiv \frac{R}{\lambda}, \tag{47}$$

where R is the outer resolved size of the system and λ is the characteristic radial scale on which the solved CPTG field organizes its curvature structure. In this sense, N measures how many effective curvature-organization lengths fit across the galaxy. Small N corresponds to a system whose curvature structure is organized on scales comparable to the system itself, while larger N corresponds to finer internal stratification of the solved field. This makes N a structural quantity, not a photometric or catalog label.

In the reduced galaxy formulation, N is extracted from the solved CPTG acceleration field $g(r)$ through the same curvature invariant that governs the theory,

$$C(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr}\right)^2. \tag{48}$$

The quantity $C(r)$ measures both field strength and field-structure, while its radial variation identifies where the solved galaxy profile is actually reorganizing its curvature support. The structural mode is therefore not imposed externally; it is read off from the solved CPTG field itself.

Operationally, we define the characteristic structural length λ from the outer galaxy, where the long-range CPTG organization is most visible. Let

$$R \equiv \max(r), \qquad w(r) \equiv \left|\frac{dC}{dr}\right|. \tag{49}$$

Restricting to the outer system $r > \frac{1}{5}R$, we define

$$\lambda = \frac{\displaystyle\int_{\frac{1}{5}R}^{R} r\, w(r)\, dr}{\displaystyle\int_{\frac{1}{5}R}^{R} w(r)\, dr}, \qquad N = \frac{R}{\lambda}. \tag{50}$$

In discrete form, this is the curvature-weighted mean outer radius of the solved profile, using the magnitude of dC/dr as the weighting field. In other words, λ is the effective outer curvature-organization scale, and N counts how many such scales fit into the full galaxy radius.

This definition explains how N works physically. If the curvature structure changes only on broad scales, then λ is large and N remains small. If the solved field develops more tightly organized radial structure, then λ decreases and N increases. The mode therefore provides a compact summary of the internal structural state of the CPTG solution, rather than merely its total mass, size, or asymptotic velocity. Because it is derived from $C(r)$, it remains tied to the same nonlinear geometric content that drives the rest of the theory.

In the present formulation, N is used as a Curvature-Weighted Structural Mode Index and can be mapped to empirical mode intervals for interpretation. Those intervals run from low-mode dwarf and irregular systems through intermediate late spirals to higher-mode early-disk and outlier classes. This mapping is based only on the derived mode value and is explicitly not intended as a replacement for catalog morphology; rather, it is a structural readout of the CPTG solution.

The advantage of introducing N is that it gives the theory a dimensionless structural diagnostic derived from its own solved field. It provides a way to compare galaxies by how curvature support is organized, not merely by luminosity class or visible morphology. In that sense, N is the natural structural companion to the galaxy-limit CPTG field equation: the field equation determines the galaxy solution, and the mode N summarizes the curvature organization of that solution.

16 CPTG Reduction for Dissociative Cluster Mergers

To describe cluster mergers within the same CPTG framework, we again begin from the master action,

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + L_m + \beta a_\star^{2/3} C^{1/3} + L_{\text{trans}} \right], \tag{51}$$

with curvature invariant

$$C = g^2 + \kappa^2 (\nabla g)^2. \tag{52}$$

The same invariant that controls the galaxy limit also controls the merger limit. The difference is that cluster mergers are dynamically anisotropic: collisional gas, collisionless galaxies, and the lensing field need not remain spatially coincident. In this regime, curvature polarization supplies broad support, while curvature transport organizes and displaces coherent curvature structure.

16.1 General Merger-Plane Reduction

In the reduced merger-plane description, the baryonic system is separated into gas and galaxy sectors. The gas traces collisional plasma, while the galaxies trace the effectively collisionless baryonic concentrations. These components define two background fields:

$$\phi_{\text{gal}}, \qquad \phi_{\text{bary}} = \phi_{\text{gal}} + \phi_{\text{gas}}. \tag{53}$$

The galaxy-dominated field defines a curvature-organization background χ_{bg}, while the full baryonic field defines a static polarization-supported background Σ_{pol}. The role of χ_{bg} is to guide the direction and coherence of transported curvature; the role of Σ_{pol} is to provide the broad lensing support produced by the nonlinear polarization response.

The merger reduction therefore contains three physical pieces:

1. a baryonic geometry supplied by gas and galaxies,
2. a broad polarization-supported lensing background $\Sigma_{\rm pol}$,
3. an explicitly transported curvature mode T_+ that carries displaced coherent curvature structure.

This construction is not a dark-matter halo prescription. The displaced lensing field arises from the response of curvature to baryonic geometry, together with directional curvature transport.

16.2 Transported-Curvature Mode

The active transported component is represented by a positive transported-curvature mode $T_+(x, y, t)$. Since the full relativistic metric is not evolved explicitly in the reduced merger plane, the transported channel is closed through the local invariant

$$C_T = T_+^2 + \kappa_T^2 |\nabla T_+|^2, \tag{54}$$

which mirrors the full CPTG invariant $C = g^2 + \kappa^2 (\nabla g)^2$.

The corresponding transported-curvature polarization field is

$$P_T = \beta C_T^{-2/3} \nabla C_T, \tag{55}$$

so that positive transported-curvature reinforcement is sourced by the positive part of $\nabla \cdot P_T$. The reduced transport equation can be written schematically as

$$\partial_t T_+ = D_{\rm trans} \nabla^2 T_+ - \mathbf{u}_{\rm eff} \cdot \nabla T_+ - \Gamma_{\rm trans} T_+ + S_+[T_+] + \mathcal{S}_{\rm CPTG}[T_+; \mathbf{u}_{\rm eff}]. \tag{56}$$

Here $D_{\rm trans}$ represents local spreading of transported curvature, $\mathbf{u}_{\rm eff}$ is the effective merger-plane transport field, $\Gamma_{\rm trans}$ suppresses unsupported structure, S_+ is the positive polarization-reinforcement source, and $\mathcal{S}_{\rm CPTG}$ is the curvature-support term.

The support term is the reduced expression of curvature persistence. It favors transported structures that are strong, gradient-supported, and coherently aligned with the transport direction. A representative reduced form is

$$\mathcal{S}_{\rm CPTG} \propto \widehat{C}_T^{1/3} (1 + \mathcal{C}_{\rm coh})^{1/3} \widehat{G}_T^{7/20} T_+, \tag{57}$$

where $\widehat{C}_T$ is the normalized transported-curvature invariant, $\widehat{G}_T$ is the normalized gradient strength, and

$$\mathcal{C}_{\rm coh} = \max\Big(0, \widehat{\nabla C_T} \cdot \hat{\mathbf{u}}_{\rm eff}\Big)^2 \tag{58}$$

measures alignment between the transported-curvature gradient and the effective transport direction.

16.3 Curvature-Weighted Lensing Projection

The observable reduced lensing field is built from a broad polarization background and a transported-curvature contribution:

$$\Sigma_{\text{final}} = B_{\text{pol}}\Sigma_{\text{pol}} + \alpha_{T,\text{eff}}\Sigma_T. \tag{59}$$

Here $B_{\text{pol}}\Sigma_{\text{pol}}$ is the static polarization-supported component, while Σ_T is the transported-curvature contribution built from T_+, $|\nabla T_+|$, and $C_T^{1/3}$. The coefficient $\alpha_{T,\text{eff}}$ is not an independent dark component; it represents the effective weight of the active transported-curvature region in the reduced projection.

Thus the lensing morphology is controlled by the combination

$$\begin{array}{r} \text{baryonic geometry} \\ \downarrow \\ \text{curvature polarization} \\ \downarrow \\ \text{directional curvature transport} \\ \downarrow \\ \text{observable lensing projection} \end{array} \tag{60}$$

16.4 Physical Interpretation

In a dissociative cluster merger, the gas is slowed and displaced by collisional plasma dynamics, while the galaxy concentrations pass through more freely. In CPTG, the lensing field does not have to remain coincident with the gas, because the observable lensing structure is not identified only with baryonic density. It is identified with the curvature response generated by baryonic geometry.

The broad part of the response is supplied by curvature polarization. The compact displaced part is supplied by transported curvature. When the transported mode remains coherent, the lensing structure can remain aligned with the collisionless galaxy distribution while separating from the collisional gas. This provides a natural CPTG interpretation of gas–lensing dissociation without introducing a separate dark matter halo.

This reasoning applies to dissociative cluster mergers generally. The Bullet Cluster is treated as the primary benchmark application because it provides a particularly clear observed separation between gas, galaxies, and lensing structure.

16.5 Bullet Cluster Benchmark Application

The Bullet Cluster is the cleanest observational test of the merger-limit mechanism because its gas, galaxy, and lensing components are strongly separated. In

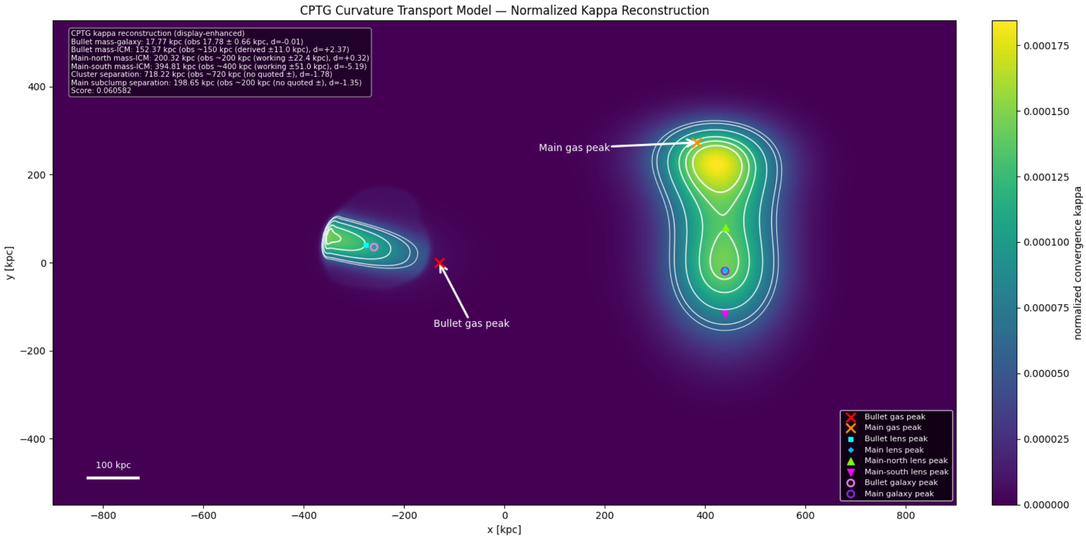

Figure 2: Reduced CPTG merger-limit reconstruction of the Bullet Cluster shown as a normalized convergence (κ) map. The image displays the broad polarization-supported background together with the localized transported-curvature structure that generates the displaced lensing morphology. The Bullet-side lensing peak remains closely aligned with the Bullet galaxy concentration while strongly separated from the collisional gas peak, and the main-cluster side resolves into north and south lensing substructure. The inset benchmark values show that the reconstructed offsets closely match the targeted Bullet Cluster observables, including the Bullet mass–galaxy offset, Bullet mass–ICM offset, main-north and main-south mass–ICM offsets, cluster-scale separation, and main-subclump separation. This figure illustrates the central merger-scale CPTG claim: the observed dissociation pattern can arise from baryonic geometry, curvature polarization, and directional curvature transport without introducing a dark matter halo.

the reduced CPTG interpretation, the Bullet-side lensing peak arises from the combination of broad polarization support and a localized transported-curvature structure. The main-cluster side can also resolve into multiple curvature-supported substructures, reflecting the nontrivial geometry of the merger.

The relevant observables are therefore not merely the presence of a gas–mass separation, but the full dissociation pattern:

- Bullet mass–galaxy offset,
- Bullet mass–ICM offset,
- main-cluster north and south mass–ICM offsets,
- cluster-scale lensing separation,
- main-subclump separation.

Matching these quantities tests whether curvature polarization and directional curvature transport can reproduce the observed merger morphology within one geometric framework.

16.6 Summary of the Reduced Merger Model

The reduced merger model is therefore characterized by:

- baryonic gas and galaxy geometry,
- a curvature-organization background χ_{bg},
- static polarization support Σ_{pol},
- positive transported-curvature evolution T_+,
- source reinforcement through $P_T = \beta C_T^{-2/3} \nabla C_T$,
- coherence-weighted curvature persistence,
- final lensing projection through Σ_{final}.

This gives the CPTG merger-limit reduction used for dissociative cluster phenomenology, while remaining explicitly a reduced merger-plane formulation rather than the unreduced relativistic field equation itself.

17 Predictions of CPTG

The Curvature Polarization Transport Gravity (CPTG) framework yields a set of concrete, testable predictions across multiple astrophysical regimes. These predictions arise from the same unified geometric mechanism: baryonic matter sources curvature, curvature develops a nonlinear polarized response, and organized curvature can in appropriate regimes be transported anisotropically. The predicted phenomena therefore do not rely on the introduction of non-baryonic matter components.

17.1 Recovery of Newtonian Gravity in Strong-Field Regimes

In regions where the gravitational acceleration satisfies

$$g \gg a_\star, \tag{61}$$

the nonlinear polarization and transport contributions become subdominant relative to the baryonic term, so that the effective gravitational response approaches

$$g \approx g_{\mathrm{bar}}, \tag{62}$$

and the theory recovers standard Newtonian behavior. Compact, strongly bound systems such as stars and inner galactic regions are therefore predicted to obey conventional gravitational dynamics to a very good approximation.

17.2 Enhanced Gravitational Response in Low-Acceleration Systems

In low-acceleration environments,

$$g \ll a_\star, \tag{63}$$

the nonlinear response becomes significant. In the bound, quasi-static galaxy regime this enhancement is predicted to be dominated by curvature polarization, while transport survives only through its coarse-grained geometric radial remnant. The effective response is then governed asymptotically by

$$g \sim \sqrt{a_\star \, g_{\rm bar}} + g_{\rm geom}(r), \tag{64}$$

leading to an enhancement relative to the baryonic contribution.

This scaling is asymptotic and is verified numerically from the implicit galaxy equation rather than obtained as a closed-form analytic solution.

This behavior predicts that galaxy rotation curves remain approximately flat through the observed weak-field regime, while deeper extrapolation under the fixed-outer-source continuation rule approaches the farther CPTG outer attractor.

17.3 Dependence on Structural Properties

Unlike purely baryonic or halo-based models, CPTG predicts that the effective gravitational response depends on the spatial structure of the system. In the full theory, this dependence is governed by curvature invariants of the form

$$C = g_j g_j + \kappa^2 (\partial_i g_j)(\partial_i g_j). \tag{65}$$

Here indices i, j denote spatial components in Cartesian coordinates, with repeated indices summed.

As a result, systems with similar baryonic mass distributions but different spatial organization can exhibit different effective gravitational behavior, even when their total baryonic content is comparable. In particular, more diffuse systems are expected to show stronger deviations from Newtonian predictions [7] because the curvature-gradient contribution becomes comparatively more important.

17.4 Directional Lensing Offsets in Cluster Mergers

In dynamically evolving systems such as merging galaxy clusters, transport contributes explicitly to the organization of the effective lensing curvature. At the effective merger level this contribution can be represented by anisotropic terms of the form

$$\mathcal{K}_{\rm lens} = \nabla^2 \Phi - s \, \nabla \cdot \left(\Sigma^2 \, \mathbf{S} \, (\mathbf{S} \cdot \nabla) \Phi \right). \tag{66}$$

The first term corresponds to the baryonic mass distribution through $\nabla^2\Phi = 4\pi G\rho$.

The transport contribution predicts that lensing peaks need not remain point-wise coincident with the collisional gas distribution. Instead, organized curvature can be guided along preferred directions defined by the transport field $\mathbf{S}$, producing offsets between the gas component, the galaxy concentrations, and the lensing signal. The magnitude and direction of these offsets are predicted to depend on merger geometry, dynamical state, and local curvature structure.

17.5 Regime-Dependent Effective Acceleration Scale

The CPTG framework introduces a characteristic acceleration scale through the nonlinear response of the curvature field. However, when evaluated in astrophysical systems, the effective value inferred from observations is predicted to depend on polarization strength, transport effects, and coarse-grained structure.

Consequently, CPTG does not require that a single universal acceleration scale be directly observed across all systems. Instead, different astrophysical regimes are expected to probe different effective realizations of the same underlying curvature dynamics.

17.6 Link Between Galactic Dynamics and Cosmological Scale

Although the effective acceleration scale may vary across systems, CPTG predicts that it is ultimately tied to the underlying curvature dynamics of the theory. This establishes a connection between galactic dynamics and the large-scale structure of spacetime.

In particular, the theory implies a relation of the form

$$H_0 = \frac{16\pi}{3}\frac{a_\star}{c}. \tag{67}$$

This relation is interpreted not as an independent postulate, but as a structural link between the characteristic acceleration scale governing low-acceleration dynamics and the cosmological curvature scale entering the broader theory.

17.7 Unified Description Across Scales

Taken together, these results imply that different astrophysical systems probe different limits of the same underlying theory:

- compact systems probe the Newtonian limit,

- galaxies probe the nonlinear polarization-dominated regime with a residual geometric transport correction,
- cluster mergers probe the regime in which directional transport and polarization both become observationally significant.

Thus CPTG predicts that phenomena commonly attributed to distinct physical mechanisms can instead be understood as regime-dependent manifestations of a single geometric framework.

18 Comparison with MOND and Emergent Acceleration Scales

Modified Newtonian Dynamics (MOND) [6] introduces a characteristic acceleration scale a_0 below which the gravitational response transitions from Newtonian behavior to an enhanced regime. In its standard formulation, MOND posits a universal constant $a_0 \sim 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, inserted phenomenologically to reproduce galaxy rotation curves.

CPTG reaches a superficially similar low-acceleration behavior by a different route. The transition does not begin from an interpolation rule between Newtonian and non-Newtonian regimes. It arises from the nonlinear curvature response encoded in the invariant

$$C = g_j g_j + \kappa^2 (\partial_i g_j)(\partial_i g_j), \tag{68}$$

together with the characteristic scale

$$a_\star = \frac{3}{16\pi} c H_0. \tag{69}$$

In this sense, MOND and CPTG may share part of their phenomenology in galaxies while differing in physical origin: MOND postulates an acceleration law, whereas CPTG attributes the departure from Newtonian gravity to curvature polarization, with transport surviving in the galaxy limit only through an effective coarse-grained geometric remnant.

18.1 Emergent Low-Acceleration Behavior

In the low-acceleration regime $g \ll a_\star$, the nonlinear response becomes significant. In the galaxy limit this response is dominated by curvature polarization, while transport enters only through the effective geometric correction $g_{\mathrm{geom}}(r)$. The reduced galaxy equation therefore yields the asymptotic behavior

$$g \sim \sqrt{a_\star\, g_{\mathrm{bar}}} + g_{\mathrm{geom}}(r), \tag{70}$$

which reproduces the characteristic MOND-like square-root scaling in the intermediate weak-field regime, while allowing the outermost behavior to retain an additional structure-sensitive geometric contribution.

In CPTG this scaling is not imposed through an interpolation function. It appears as a limiting behavior of the nonlinear field equation and depends not only on the local field strength but also on the structural content of the field through the invariant C.

18.2 Absence of a Universal Acceleration Constant

A central difference from MOND is that CPTG does not predict a strictly universal phenomenological transition scale for all galaxies. While $a_\star$ provides a fundamental scale tied to cosmology, the effective onset of the nonlinear response depends on additional geometric and structural factors, including:

- the underlying curvature structure of the field,
- the gradient structure of the acceleration field,
- the effective geometric correction term g_{geom},
- and the coarse-grained remnant of transport in the bound-system limit.

As a result, two galaxies with similar baryonic masses need not enter the nonlinear regime in exactly the same way if their internal structural organization differs. CPTG therefore provides a natural geometric origin for mild scatter around idealized scaling relations without requiring extra phenomenological freedom to be inserted by hand.

18.3 Improved Flexibility in Galaxy Fits

Because CPTG depends on both local field strength and spatial structure, it can accommodate variations in rotation-curve shape that are difficult to capture with a single MOND interpolation function. In particular:

- inner galaxy regions can transition more gradually depending on the local gradient structure,
- outer regions can receive additional contributions from the geometric correction term,
- and departures from idealized scaling relations can arise naturally from the underlying curvature dynamics.

This additional flexibility does not come from introducing an arbitrary second law for galaxies. It comes from allowing the same underlying invariant structure to respond differently in systems with different geometric organization.

18.4 Connection to Cosmology

In CPTG, the characteristic acceleration scale is not arbitrary but linked to cosmological expansion through

$$a_\star = \frac{3}{16\pi} c H_0. \tag{71}$$

The corresponding structural length scale is not introduced independently but is derived from the same relation. Eliminating H_0 yields

$$R_c = \frac{48\pi c^2}{a_\star}, \tag{72}$$

so that the acceleration scale and the structural scale arise from a single underlying curvature scale.

This establishes a direct structural connection between galaxy dynamics and large-scale cosmology. In MOND, by contrast, the numerical proximity of a_0 to cH_0 is often regarded as suggestive but external to the theory. In CPTG the corresponding relation is built into the scale-setting structure of the framework itself.

18.5 Summary of Differences

Feature	MOND	CPTG
Acceleration scale	Fixed a_0	Emergent $a_\star$
Scaling origin	Interpolation	Nonlinear field equation
Structure	Minimal	Explicit field structure
Cosmology link	Empirical	Derived relation
Transport role	None	Geometric remnant
Flexibility	Limited	Higher

CPTG therefore retains the successful low-acceleration phenomenology usually associated with MOND while replacing a universal phenomenological constant with a structurally generated nonlinear response. In that sense, MOND can be viewed as capturing part of the galaxy-scale phenomenology that CPTG seeks to derive from a deeper geometric mechanism.

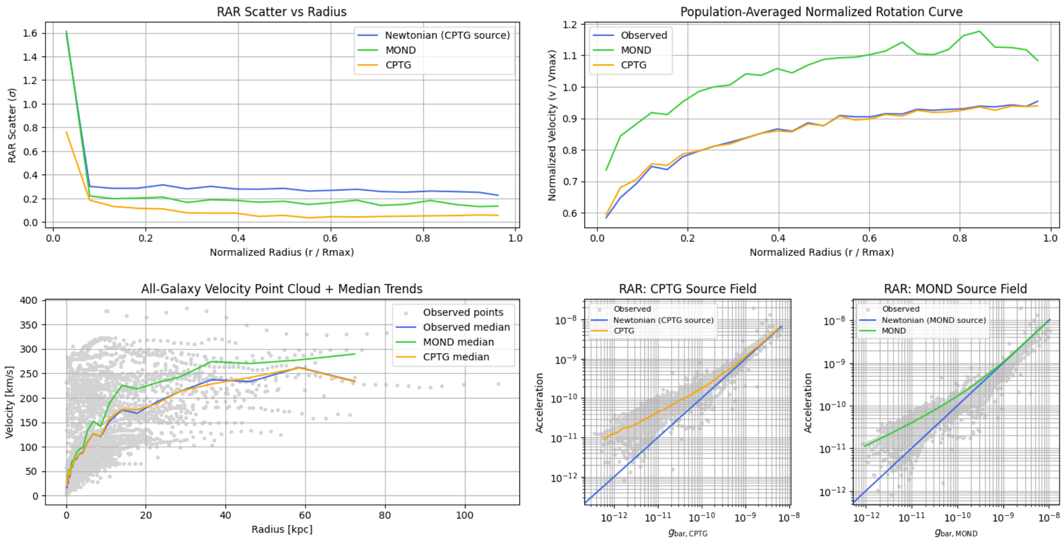

Figure 3: Composite CPTG–MOND benchmark comparison on the SPARC galaxy sample, showing stacked and population-averaged diagnostics for all 175 galaxies. **Top left:** RAR scatter as a function of normalized radius r/R_{max}, showing that CPTG maintains the lowest scatter across most of the disk, below both the Newtonian baseline computed from the CPTG source field and MOND. **Top right:** population-averaged normalized rotation curve, where the CPTG prediction closely follows the observed mean profile, while MOND systematically overshoots the normalized velocity, especially toward larger radii. **Bottom left:** all-galaxy velocity point cloud with median trends, showing the observed SPARC velocity ensemble together with the median CPTG and MOND predictions as a function of physical radius. **Bottom middle:** radial acceleration relation using the CPTG source field, plotting the observed acceleration cloud against $g_{\mathrm{bar,CPTG}}$, with the Newtonian reference and the CPTG trend overlaid. **Bottom right:** radial acceleration relation using the MOND source field, plotting the observed acceleration cloud against $g_{\mathrm{bar,MOND}}$, with the Newtonian reference and the MOND trend overlaid. Taken together, these diagnostics show that CPTG more closely reproduces the SPARC phenomenology than MOND in the stacked scatter behavior, the population-averaged rotation-curve profile, the velocity medians, and the acceleration-relation trends.

19 Connection to the Baryonic Tully–Fisher Relation

The baryonic Tully–Fisher relation (BTFR) is one of the most important empirical regularities in galaxy dynamics. It states that the total baryonic mass of a rotationally supported galaxy is strongly correlated with the fourth power of its characteristic outer rotation speed,

$$M_b \propto V_f^4, \tag{73}$$

or equivalently,

$$V_f^4 \propto M_b. \tag{74}$$

Here M_b denotes the baryonic mass and V_f denotes the approximately flat or outer characteristic rotation velocity. In the CPTG framework, this relation is not imposed as an empirical fitting rule. It appears as the leading polarization-dominated low-acceleration scaling of the reduced galaxy-limit equation.

19.1 CPTG Galaxy-Limit Equation

In the reduced galaxy limit, the CPTG acceleration field is governed by the implicit nonlinear relation

$$g(r) = g_{\mathrm{bar}}(r) + \frac{\sqrt{a_\star g_{\mathrm{bar}}(r)} + g_{\mathrm{geom}}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}. \tag{75}$$

The geometric transport remnant is

$$g_{\mathrm{geom}}(r) = \eta a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \qquad \eta = \frac{1}{2}, \tag{76}$$

with

$$R_c = \frac{48\pi c^2}{a_\star}. \tag{77}$$

The CPTG baryonic source field is constructed from the observed baryonic components as

$$g_{\mathrm{bar}}(r) = \frac{\frac{27}{50}v_{\mathrm{gas}}^2(r) + \frac{23}{50}v_{\mathrm{disk}}^2(r) + \frac{16}{25}v_{\mathrm{bul}}^2(r)}{r}. \tag{78}$$

This quantity is the baryonic input to the nonlinear CPTG response. It is not the final gravitational field.

19.2 Intermediate Low-Acceleration Regime

The BTFR arises in the intermediate low-acceleration regime relevant to the observed flat portions of galaxy rotation curves. In the weak-field source regime,

$$g_{\mathrm{bar}}(r) \ll a_\star, \tag{79}$$

and for the corresponding low-acceleration solution,

$$g(r) \ll a_\star, \tag{80}$$

the nonlinear suppression factor in Eq. (75) approaches unity to leading order. If

$$g_{\mathrm{bar}}(r) \ll \sqrt{a_\star g_{\mathrm{bar}}(r)} \tag{81}$$

and

$$|g_{\mathrm{geom}}(r)| \ll \sqrt{a_\star g_{\mathrm{bar}}(r)} \tag{82}$$

over the flat-curve interval, the leading CPTG response reduces to

$$g(r) \simeq \sqrt{a_\star g_{\mathrm{bar}}(r)}. \tag{83}$$

This is the key leading-order CPTG scaling behind the baryonic Tully–Fisher relation.

For a galaxy whose outer baryonic source is effectively enclosed within the radius under consideration, the CPTG-weighted baryonic acceleration may be approximated as

$$g_{\rm bar}(r) \simeq \frac{GM_{\rm eff}}{r^2}, \tag{84}$$

where $M_{\rm eff}$ is the effective enclosed source mass that would produce the same outer CPTG-weighted baryonic acceleration under the asymptotic GM/r^2 approximation. The circular acceleration is

$$g(r) = \frac{V^2(r)}{r}. \tag{85}$$

Substituting Eqs. (84) and (85) into Eq. (83) gives

$$\frac{V^2(r)}{r} \simeq \sqrt{a_\star \frac{GM_{\rm eff}}{r^2}}. \tag{86}$$

The radial dependence cancels:

$$V^2(r) \simeq \sqrt{Ga_\star M_{\rm eff}}, \tag{87}$$

and therefore

$$V_f^4 \simeq Ga_\star M_{\rm eff}. \tag{88}$$

Equation (88) is the leading CPTG form of the baryonic Tully–Fisher scaling.

19.3 Relation Between $M_{\rm eff}$ and M_b

The empirical BTFR is usually written in terms of the physical baryonic mass M_b. In the CPTG benchmark implementation, the baryonic source field is not treated as a purely unit-weight Newtonian source. Instead, the gas, disk, and bulge contributions enter through the CPTG source mapping in Eq. (78). It is therefore useful to define an effective CPTG source mass by

$$M_{\rm eff}(r) \equiv \frac{r^2 g_{\rm bar}(r)}{G}. \tag{89}$$

This definition should be understood as an effective enclosed source proxy in the outer-field approximation, not as a claim that a disk galaxy is exactly a spherical mass distribution.

For galaxies whose CPTG source mapping is approximately proportional to the ordinary baryonic mass in the outer disk, one may write

$$M_{\rm eff} = \Xi M_b, \tag{90}$$

where Ξ is a dimensionless source-structure factor determined by the relative gas, disk, and bulge contributions and by the internal organization of the baryonic field. Substituting Eq. (90) into Eq. (88) gives

$$V_f^4 \simeq Ga_\star \Xi M_b. \tag{91}$$

Thus CPTG predicts the observed fourth-power BTFR slope as a leading-order consequence of the curvature-polarization response, while allowing the normalization and residual structure to depend on the solved internal geometry of the galaxy.

19.4 Role of the Structural Scale $a_\star$

A central distinction between CPTG and phenomenological acceleration-law models is the interpretation of the acceleration scale. In CPTG, $a_\star$ is not introduced as a universal interpolation constant. It is the characteristic curvature-polarization acceleration scale of the system and is related to the cosmological scale through

$$a_\star = \frac{3}{16\pi} c H_0. \tag{92}$$

Equivalently,

$$H_0 = \frac{16\pi}{3} \frac{a_\star}{c}. \tag{93}$$

In the theory, $a_\star$ is interpreted as a structural curvature scale of the galaxy. In the present benchmark implementation, this scale is estimated for each galaxy and then used to derive R_c through Eq. (77). The BTFR normalization in CPTG is therefore controlled by a structural curvature scale rather than by an externally imposed universal constant.

Using Eq. (92), the leading CPTG BTFR relation may also be written as

$$V_f^4 \simeq G \left(\frac{3}{16\pi} c H_0 \right) M_{\rm eff}, \tag{94}$$

or, in terms of physical baryonic mass,

$$V_f^4 \simeq G \left(\frac{3}{16\pi} c H_0 \right) \Xi M_b. \tag{95}$$

This gives the BTFR a direct structural connection to the cosmological scale.

19.5 Structural Residuals and the Mode Number N

The CPTG framework does not require the BTFR to be an exact one-parameter law with completely structure-free residuals. Because the gravitational response depends on curvature structure, galaxies with similar baryonic masses can show different detailed rotation-curve behavior if their solved curvature support is organized differently.

The relevant structural diagnostic is the Curvature-Weighted Structural Mode Index,

$$N \equiv \frac{R}{\lambda}, \tag{96}$$

where R is the outer resolved size of the galaxy and λ is the curvature-weighted outer organization scale of the solved CPTG field. The mode is derived from

$$C(r) = g^2(r) + \kappa_{\rm struct}^2 \left(\frac{dg}{dr} \right)^2, \tag{97}$$

with the weighting field

$$w(r) = \left| \frac{dC}{dr} \right| . \tag{98}$$

The characteristic structural length is

$$\lambda = \frac{\int_{R/5}^{R} r\, w(r)\, dr}{\int_{R/5}^{R} w(r)\, dr}, \qquad N = \frac{R}{\lambda}. \tag{99}$$

The mode number N is therefore a structural readout of the solved CPTG field, not a catalog morphology label.

Taking logarithms relative to fixed reference units, Eq. (91) gives

$$\log M_b = 4 \log V_f - \log(G a_\star \Xi). \tag{100}$$

For residuals measured in $\log M_b$ at fixed V_f, one therefore obtains schematically

$$\Delta_{\mathrm{BTFR}}^{(M|V)} \simeq -\Delta \log a_\star - \Delta \log \Xi + \Delta_{\mathrm{geom}}, \tag{101}$$

where Δ_{geom} denotes the contribution of the geometric transport remnant and other higher-order structural effects beyond the leading polarization-dominated approximation.

Thus, at fixed V_f, residuals from the ideal BTFR ridge are expected to track variations in $a_\star$, Ξ, and the transport remnant g_{geom}. Since these quantities are tied to the solved curvature structure, CPTG predicts that BTFR residuals should not be purely random. They should show organization with structural diagnostics such as N, source composition, and outer-field curvature gradients.

19.6 Distinction from MOND

The leading CPTG relation

$$V_f^4 \simeq G a_\star M_{\mathrm{eff}} \tag{102}$$

has the same fourth-power form as the familiar low-acceleration MOND scaling. The physical interpretation, however, is different.

In MOND, the BTFR follows from imposing a universal acceleration scale a_0 and an interpolation law between the Newtonian and low-acceleration regimes. In CPTG, the square-root behavior arises as the leading low-acceleration limit of the nonlinear curvature-polarization response. The acceleration scale $a_\star$ is tied to the structural and cosmological scale of the system, and the outer behavior also contains the coarse-grained geometric transport remnant $g_{\mathrm{geom}}(r)$.

Thus MOND and CPTG can share the same leading BTFR slope while assigning it different physical origins:

$$\mathrm{MOND}: \qquad V_f^4 \simeq G a_0 M_b, \tag{103}$$

where a_0 is a fixed empirical scale, whereas

$$\mathrm{CPTG}: \qquad V_f^4 \simeq G a_\star M_{\mathrm{eff}}, \tag{104}$$

where $a_\star$ and M_{eff} are tied to the nonlinear curvature structure of the galaxy.

19.7 Observed Flat Regime Versus Asymptotic CPTG Outer Regime

The BTFR is associated with the observed flat or nearly flat portion of galaxy rotation curves. In CPTG, this corresponds primarily to the intermediate low-acceleration regime in which the polarization term produces the dominant fourth-power scaling. This should be distinguished from the farther extended outer regime of the CPTG solution.

At sufficiently large radii beyond the observed region, the fixed-outer-source continuation of the CPTG solution enters a low-acceleration regime in which the square-root polarization response controls the leading scaling. The geometric transport remnant remains part of the full solution, but the leading $V(r) \propto r^{1/4}$ behavior follows from the polarization response under this continuation rule,

$$V(r) \propto r^{1/4}, \tag{105}$$

or equivalently

$$g(r) \propto r^{-1/2}. \tag{106}$$

Therefore, CPTG predicts two related but distinct behaviors:

1. an intermediate polarization-dominated regime that produces the baryonic Tully–Fisher scaling,
2. a farther extended low-acceleration continuation regime with $V(r) \propto r^{1/4}$.

Current rotation-curve observations mostly probe the transition into the intermediate weak-field regime rather than the fully asymptotic CPTG outer attractor.

19.8 Testable CPTG Predictions for the BTFR

The CPTG interpretation of the BTFR leads to several testable predictions.

First, the dominant BTFR slope should be close to four because the polarization-dominated low-acceleration response gives

$$g \sim \sqrt{a_\star g_{\rm bar}}, \tag{107}$$

provided that the product $a_\star \Xi$ is not itself strongly mass-dependent across the sampled galaxy population.

Second, the BTFR normalization should be controlled by the effective product

$$a_\star \Xi, \tag{108}$$

rather than by a strictly universal empirical acceleration constant.

Third, residuals from the BTFR ridge should correlate with structural properties of the solved field, especially the mode number N, the outer curvature-gradient structure, and the relative gas, disk, and bulge contributions to $g_{\rm bar}$.

Fourth, galaxies in similar structural mode ranges should occupy similar residual bands around the main BTFR relation. In this sense, CPTG predicts a BTFR ridge with curvature-structured residuals rather than a purely featureless scatter distribution.

Fifth, deeper observations extending beyond the currently measured outer rotation curves should begin to distinguish the ordinary BTFR flat-regime behavior from the farther CPTG outer attractor.

19.9 Summary

CPTG naturally reproduces the baryonic Tully–Fisher scaling in the intermediate low-acceleration galaxy regime. Starting from the reduced galaxy-limit equation, the leading curvature-polarization response gives

$$g \simeq \sqrt{a_\star g_{\rm bar}}, \tag{109}$$

under the polarization-dominated conditions

$$g_{\rm bar} \ll \sqrt{a_\star g_{\rm bar}}, \qquad |g_{\rm geom}| \ll \sqrt{a_\star g_{\rm bar}}. \tag{110}$$

Using

$$g_{\rm bar} \simeq \frac{GM_{\rm eff}}{r^2}, \qquad g = \frac{V^2}{r}, \tag{111}$$

one obtains

$$V_f^4 \simeq G a_\star M_{\rm eff}. \tag{112}$$

For an effective source mass proportional to the physical baryonic mass, this becomes

$$V_f^4 \simeq G a_\star \Xi M_b. \tag{113}$$

Thus the BTFR appears in CPTG as a consequence of nonlinear curvature polarization, not as an imposed interpolation law or a dark matter halo scaling. The same framework also predicts that residuals about the BTFR should carry structural information through $a_\star$, Ξ, $g_{\rm geom}$, and the Curvature-Weighted Structural Mode Index N.

20 Comparison with ΛCDM

The standard cosmological model, ΛCDM [5], explains galaxy dynamics, cluster lensing, and large-scale structure by supplementing baryonic matter with cold dark matter (CDM) and a cosmological constant Λ. In that framework, apparent excess gravity is interpreted as the gravitational effect of an additional, nonluminous matter component. CPTG takes a fundamentally different position: the same baryonic geometry sources a nonlinear curvature response, and the resulting phenomenology is attributed to curvature polarization and curvature transport derived from a single variational principle rather than from an added matter sector.

This difference is not merely interpretive. It changes the ontology of gravitational enhancement, the relation between local dynamics and cosmology, and the way galaxy and cluster phenomena are connected to one another.

20.1 Origin of Gravitational Enhancement

In ΛCDM, the enhanced gravitational effects observed in galaxies and clusters are attributed to dark matter halos or subhalos surrounding the baryonic structures. The gravitational field is still governed by general relativity, but the source content is enlarged by a new stress-energy component. Halo structure is then described by empirical or simulation-motivated profiles such as Navarro–Frenk–White (NFW) distributions [9].

In CPTG, the enhancement arises from the gravitational field itself. Baryonic matter generates curvature, the curvature invariant

$$C = g^2 + \kappa^2(\nabla g)^2, \tag{114}$$

produces a nonlinear polarization response, and the resulting polarization field modifies the effective source relation. In dynamically evolving systems, the same framework also permits anisotropic curvature transport. The excess gravitational phenomenology is therefore attributed to nonlinear geometric response rather than to an additional matter component.

20.2 Galaxy Rotation Curves

Within ΛCDM, galaxy rotation curves are explained by embedding the observed baryons inside extended dark matter halos. The resulting curves depend on both the baryonic distribution and the assumed halo profile, which introduces degeneracies among stellar mass-to-light ratios, halo concentrations, and halo masses.

In CPTG, the galaxy regime is polarization-dominated. The rotation curve is determined directly from the baryonic distribution through the effective galaxy equation,

$$g(r) = g_{\rm bar}(r) + \frac{\sqrt{a_\star\, g_{\rm bar}(r)} + g_{\rm geom}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}. \tag{115}$$

Here the leading low-acceleration enhancement is a consequence of curvature polarization, while the transport sector survives only through the coarse-grained geometric correction $g_{\rm geom}(r)$. The gravitational response is thus tied more directly to the observed baryonic structure and to the nonlinear geometry of the field, rather than to an independent halo construction.

20.3 Cluster Lensing and Mergers

In merging galaxy clusters, ΛCDM attributes the observed separation between collisional gas and lensing peaks to collisionless dark matter passing through the merger with little drag, while the gas is slowed hydrodynamically.

In CPTG, the same observational pattern is interpreted differently. The broad lensing support arises from curvature polarization, while the displaced and organized lensing structure arises from curvature transport. The merger phenomenology is therefore not described as baryons separating from an unseen collisionless matter component, but as baryonic geometry generating a polarization-supported curvature field whose organized structure can become directionally redistributed during the merger. In this sense, cluster offsets are predicted to trace transported curvature rather than transported dark matter.

20.4 Parameterization and Predictive Structure

ΛCDM relies on multiple effective ingredients at different scales, including:

- dark matter halo profiles,
- concentration and assembly-history relations,
- baryonic feedback prescriptions.

These ingredients can vary significantly from system to system and are often constrained statistically or phenomenologically.

CPTG, by contrast, is organized around a smaller set of action-level quantities,

- polarization coupling β,
- curvature scale κ,
- transport coupling λ_t,
- structural acceleration scale $a_\star$.

The effective behavior of a given system is then determined by the nonlinear structure of the equations and by the geometry of the baryonic source, not by assigning that system an independent dark halo with its own free profile.

20.5 Connection to Cosmology

In ΛCDM, the cosmological constant Λ drives late-time cosmic acceleration, while dark matter governs structure formation. The relation between galaxy-scale dynamics and cosmology is therefore indirect and mediated by formation history, halo growth, and environmental evolution.

In CPTG, the characteristic acceleration scale is linked directly to cosmology through

$$a_\star = \frac{3}{16\pi} c H_0, \tag{116}$$

so the weak-field structural response is connected to an effective expansion scale associated with the large-scale curvature structure of the system. In this sense, the relation between local dynamics and cosmology is built into the theory at the level of the characteristic scale rather than inherited indirectly through halo formation.

20.6 Summary of Differences

Feature	ΛCDM	CPTG
Gravity source	CDM halo	Curvature response
Galaxy regime	Baryons in halos	Polarization field
Cluster regime	Collisionless DM	Polarization + transport
Structure	Halo-profile based	Geometry/invariant based
Acceleration scale	Cosmology-linked	H_0-linked
Theory type	Matter extension (GR)	Modified gravity

CPTG therefore offers an alternative physical interpretation of the phenomena ordinarily assigned to dark matter in ΛCDM. The theory does not deny the observational facts that motivate the standard model; rather, it attributes the same galaxy-scale and cluster-scale anomalies to nonlinear curvature polarization and transport within a single geometric framework.

21 Implementation Summary

21.1 Galaxy Reduction Summary

For bound galactic systems, the reduced CPTG description proceeds as follows:

1. Construct the baryonic acceleration field $g_{\mathrm{bar}}(r)$ from the visible matter distribution, as in Eq. (28).
2. Treat $g_{\mathrm{bar}}(r)$ as the baryonic input to the nonlinear CPTG response rather than as the final gravitational field.
3. Solve the implicit galaxy-limit relation Eq. (24), in which curvature polarization provides the dominant low-acceleration enhancement while transport survives only through its coarse-grained geometric remnant $g_{\mathrm{geom}}(r)$ from Eq. (25).
4. Recover the circular velocity through $v(r) = \sqrt{g(r)\, r}$.

21.2 Cluster-Merger Reduction Summary

For dynamically evolving cluster mergers, the reduced CPTG description proceeds as follows:

1. Construct the galaxy-dominated background proxy ϕ_{gal} and the full baryonic background proxy ϕ_{bary}, as summarized in the general merger-plane reduction.
2. Build the static polarization-supported background Σ_{pol} from the nonlinear curvature response of the baryonic geometry.
3. Evolve the positive transported-curvature mode T_+ through the reduced merger equation in Eq. (56), with transported-curvature invariant C_T defined in Eq. (54), source reinforcement from Eq. (55), and coherence-weighted support from Eq. (57).
4. Construct the final observable lensing field by combining the polarization-supported background and transported-curvature contribution according to Eq. (59).

22 Discussion

CPTG presents galaxy dynamics and cluster-merger lensing within a single geometric framework rather than treating them as separate phenomenological problems. The central claim is not merely that gravity is strengthened in some regimes, but that curvature can respond nonlinearly through polarization and can also be organized directionally through transport. These two effects are related consequences of the same underlying framework, but they play different roles in different astrophysical environments.

In bound galactic systems, the dominant visible effect is curvature polarization. The field responds not only to baryonic source strength but also to the local geometric structure encoded in the invariant C, so the effective gravitational response is enhanced in a controlled way in low-acceleration regions while remaining Newtonian in the strong-field limit. Transport is not absent in this regime, but it survives only as a coarse-grained geometric remnant inside the effective radial reduction.

In cluster mergers, by contrast, directional transport becomes explicit. The theory then allows organized curvature structure to be guided, displaced, and maintained relative to the collisional gas distribution. In this regime, the observed lensing morphology is interpreted as the result of broad polarization support together with transported-curvature structure, rather than as evidence for an additional collisionless dark matter component.

This viewpoint has an important conceptual consequence. The characteristic scales that appear phenomenologically need not be treated as universally fixed inputs imposed from outside the theory. Because the response depends on structure, gradients, and dynamical organization, quantities inferred from data may instead reflect different realizations of the same underlying nonlinear geometric law.

The principal strength of CPTG is therefore its unifying ontology. Compact systems probe the Newtonian limit, galaxies probe the polarization-dominated nonlinear regime, and cluster mergers probe the regime in which polarization and directional transport act together to produce displaced lensing structure. The same framework is intended to account for all three without enlarging the matter sector.

23 Conclusion

We have developed Curvature Polarization Transport Gravity (CPTG) as a nonlinear extension of gravity defined by a single variational principle. Starting from a master action containing both curvature-polarization and transport sectors, we derived modified field equations in which the gravitational response depends on both the magnitude of the field and its spatial structure. The central claim of the theory is therefore not merely that gravity is strengthened in selected regimes, but that curvature can respond nonlinearly through polarization and can also be organized directionally through transport within one unified framework.

In the quasi-static galactic limit, the theory reduces to an effective coarse-grained relation in which the dominant visible effect is curvature polarization. This reduction recovers Newtonian behavior in high-acceleration regimes while yielding an enhanced gravitational response in low-acceleration regions. In this regime, transport is not absent, but survives only as a coarse-grained geometric remnant inside the effective radial dynamics. As a result, the observed flattening of galaxy rotation curves emerges without the introduction of non-baryonic matter.

In dynamically evolving systems such as cluster mergers, the transport sector becomes explicit. The theory then permits organized curvature structure to be guided, displaced, and maintained relative to the collisional baryonic plasma. The corresponding lensing morphology is interpreted as the result of broad polarization support together with transported-curvature structure, rather than as evidence for an additional collisionless dark matter component.

Taken together, these results show that CPTG is intended as a unified description of multiple gravitational regimes. Compact systems probe the Newtonian limit, galaxies probe the polarization-dominated nonlinear regime, and cluster mergers probe the regime in which polarization and directional transport act together to produce displaced lensing structure. The same geometric framework is intended to account for all three without enlarging the matter sector.

An important consequence of CPTG is that effective phenomenological scales need not be treated as universally fixed inputs imposed from outside the theory. Because the response depends on structure, gradients, and dynamical organization, quantities inferred from observations, such as characteristic acceleration scales or effective expansion rates, may instead reflect different realizations of the same underlying nonlinear geometric law.

The theory makes clear, testable predictions. It requires Newtonian recovery in strong-field systems, enhanced gravitational response in low-acceleration galaxies, and directional lensing offsets in merging clusters. These predictions provide a direct pathway for observational validation or falsification.

CPTG therefore offers a geometric alternative in which phenomena commonly attributed to dark matter arise from nonlinear curvature polarization and curvature transport. Further work will focus on detailed observational tests, development of the full field equations beyond the reduced limits presented here, and extension of the framework to fully relativistic and cosmological settings.

References

[1] D. Hilbert, "Die Grundlagen der Physik," *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 395–407 (1915).

[2] F. Zwicky, "Die Rotverschiebung von extragalaktischen Nebeln," *Helvetica Physica Acta* **6**, 110 (1933).

[3] V. C. Rubin, N. Thonnard, and W. K. Ford, "Rotational properties of 21 SC galaxies," *Astrophys. J.* **238**, 471 (1980).

[4] D. Clowe *et al.*, "A direct empirical proof of the existence of dark matter," *Astrophys. J. Lett.* **648**, L109 (2006).

[5] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).

[6] M. Milgrom, "A modification of the Newtonian dynamics," *Astrophys. J.* **270**, 365 (1983).

[7] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial acceleration relation in rotationally supported galaxies," *Phys. Rev. Lett.* **117**, 201101 (2016).

[8] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass models for 175 disk galaxies," *Astron. J.* **152**, 157 (2016).

[9] J. F. Navarro, C. S. Frenk, and S. D. M. White, "A universal density profile from hierarchical clustering," *Astrophys. J.* **490**, 493 (1997).

[10] A. Einstein, "The field equations of gravitation," *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1915).

[11] CPTG, *Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity*, companion resource for the present work, available at `https://github.com/CLG2025/CPTG`